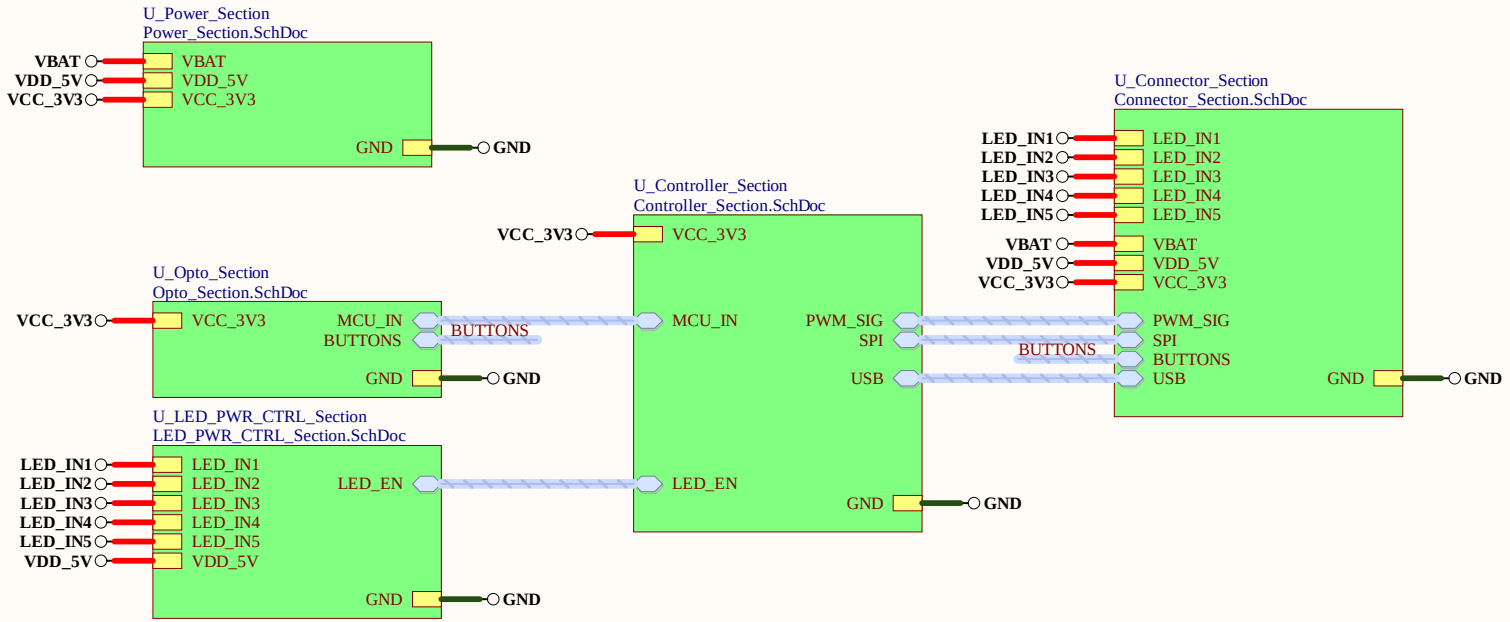
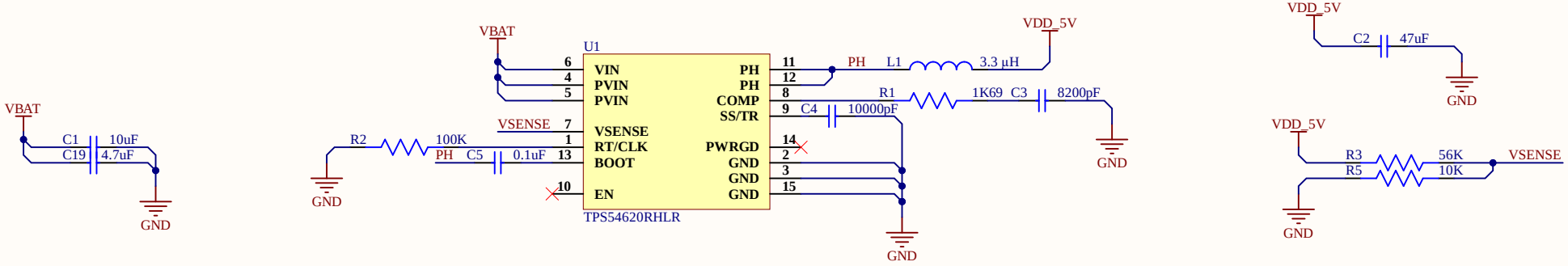
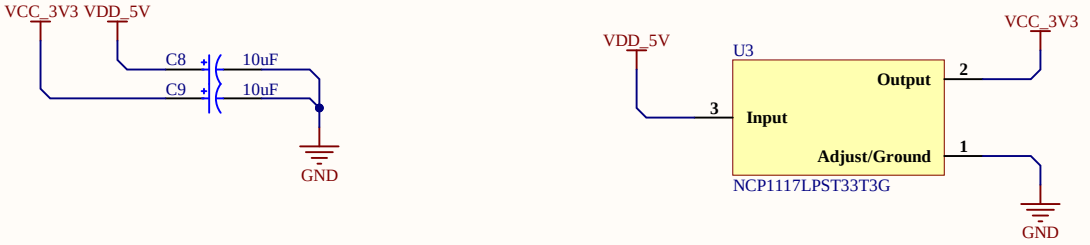
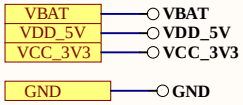

U_Power_Section
Power_Section.SchDoc
VBAT
VDD_5V
VCC_3V3
GND
GND
U_Opto_Section
Opto_Section.SchDoc
VCC_3V3
MCU_IN
BUTTONS
GND
GND
U_LED_PWR_CTRL_Section
LED_PWR_CTRL_Section.SchDoc
LED_IN1
LED_IN2
LED_IN3
LED_IN4
LED_IN5
VDD_5V
LED_EN
GND
GND
U_Controller_Section
Controller_Section.SchDoc
VCC_3V3
MCU_IN
PWM_SIG
SPI
USB
LED_EN
GND
GND
U_Connector_Section
Connector_Section.SchDoc
LED_IN1
LED_IN2
LED_IN3
LED_IN4
LED_IN5
VBAT
VDD_5V
VCC_3V3
PWM_SIG
SPI
BUTTONS
USB
GND
GND

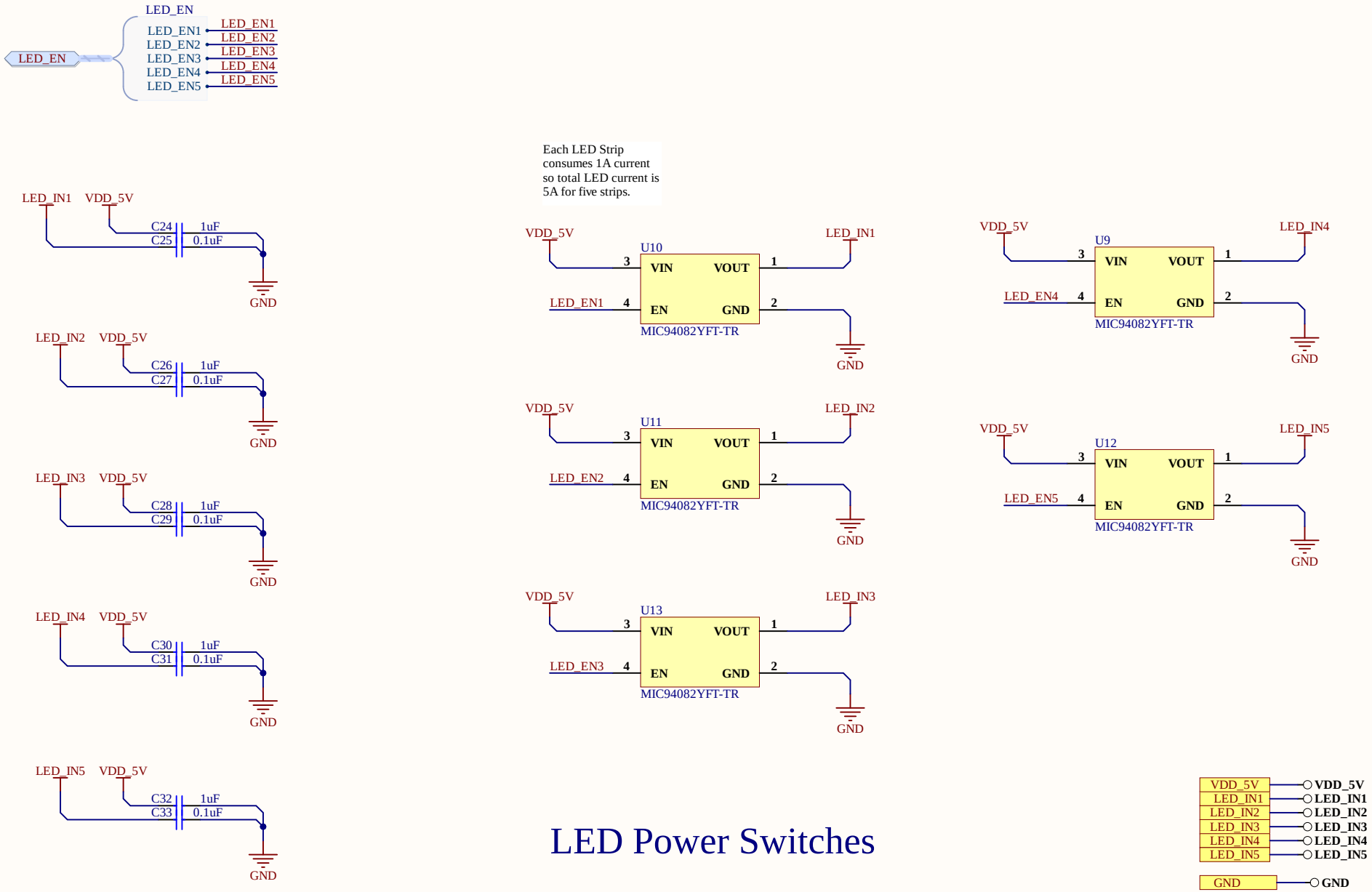


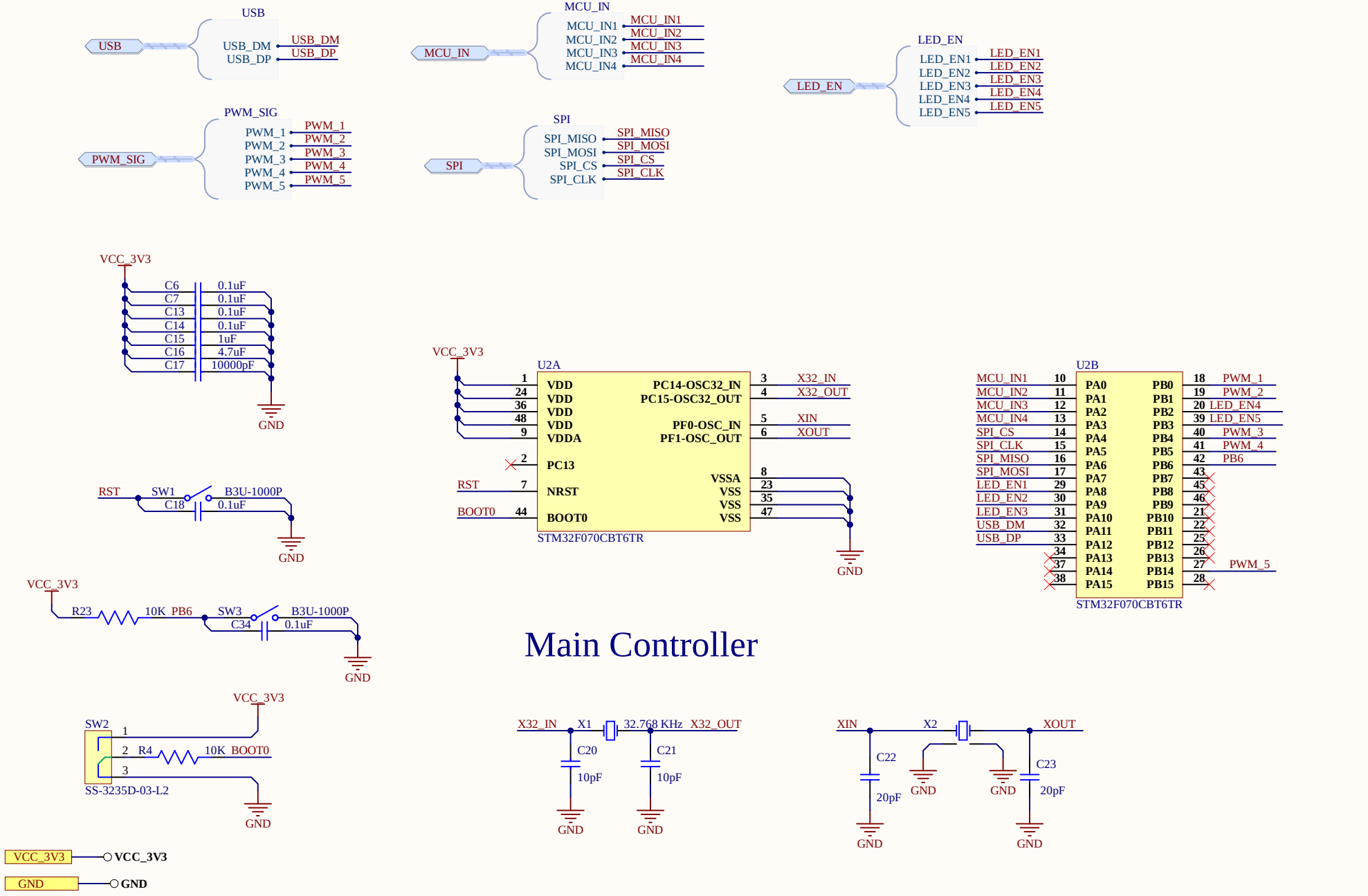
LED Power

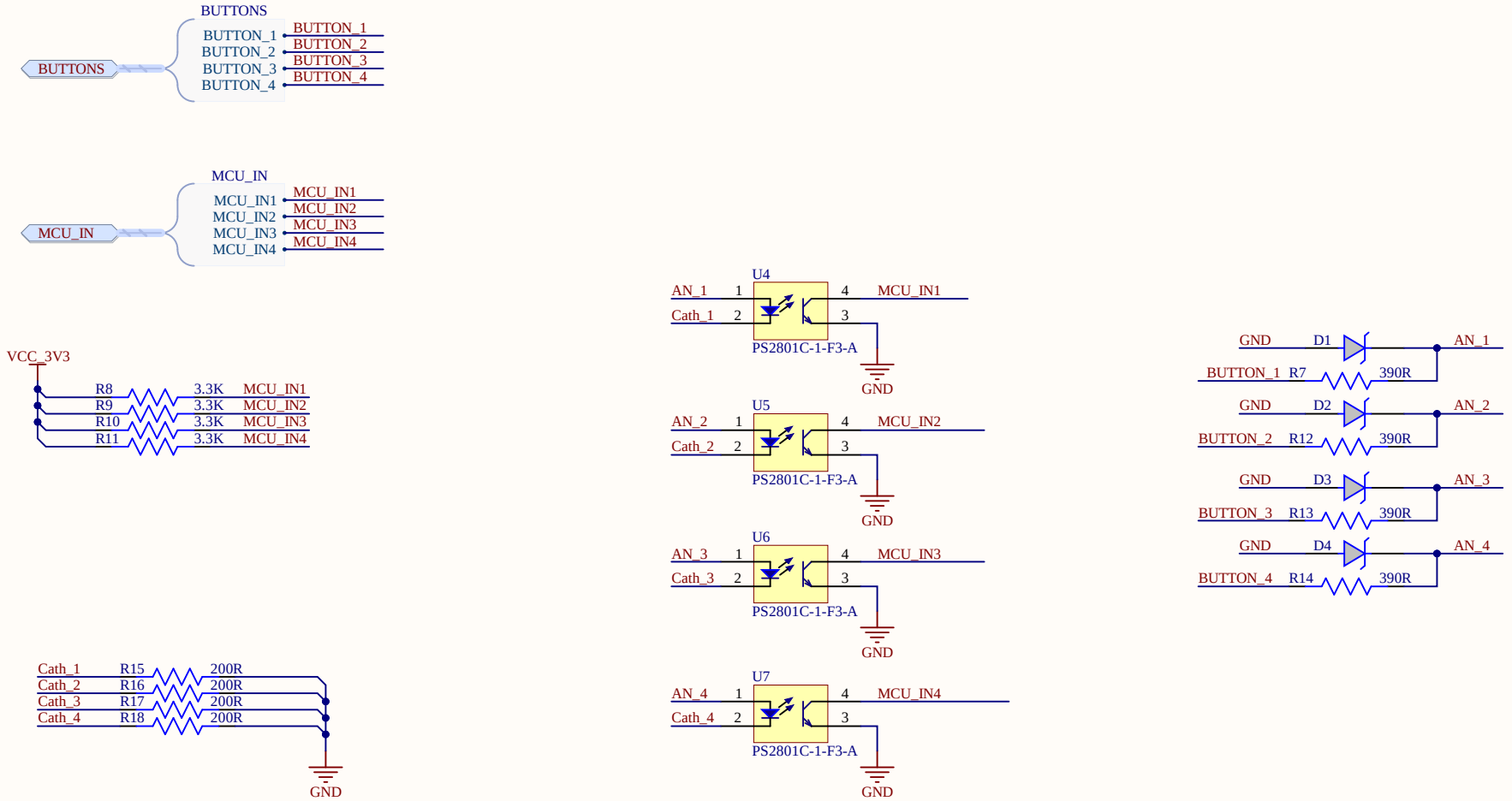


MCU Power

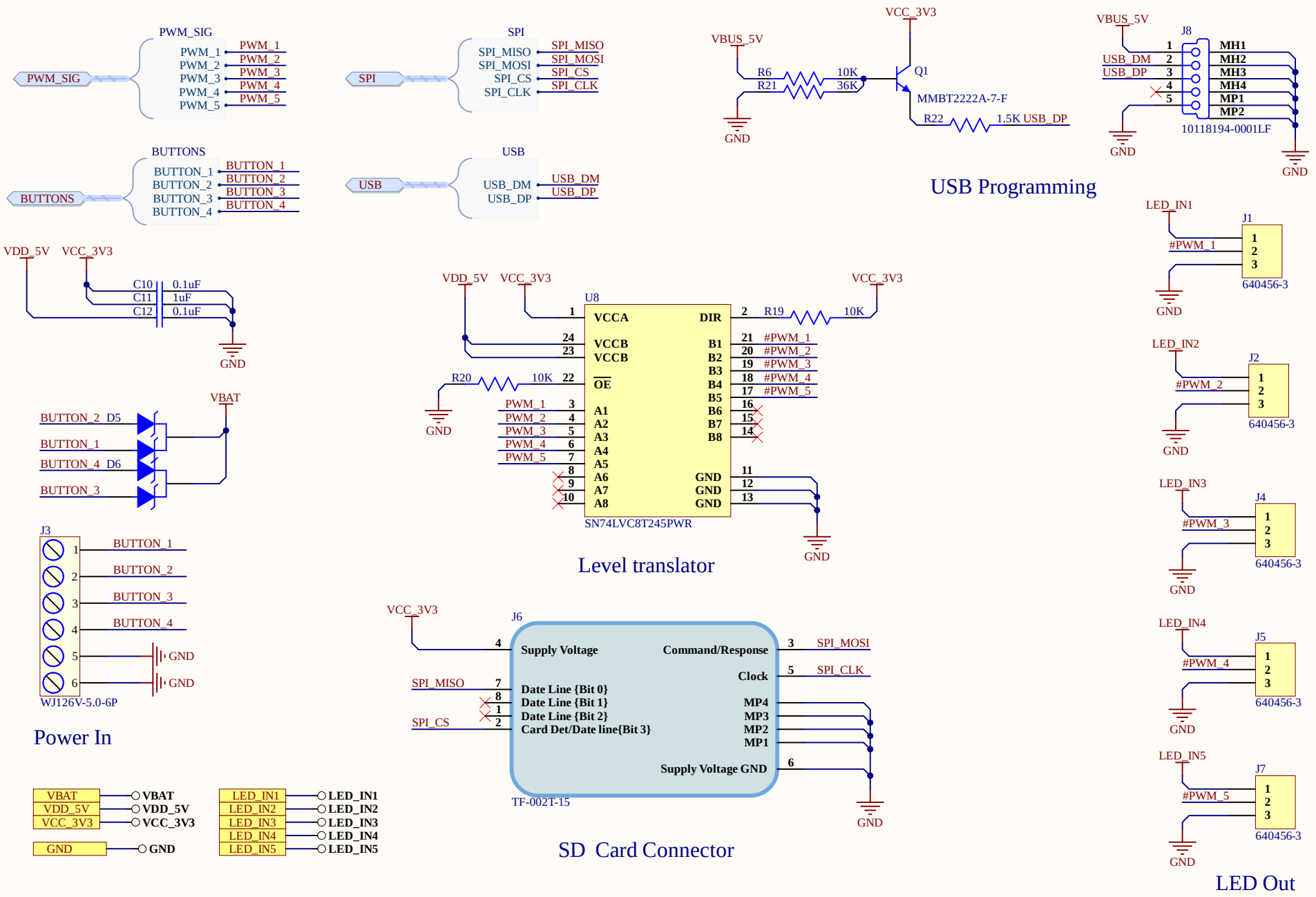



LED_EN
LED_EN
LED_EN1
LED_EN2
LED_EN3
LED_EN4
LED_EN5
LED_EN1
LED_EN2
LED_EN3
LED_EN4
LED_EN5
Each LED Strip consumes 1A current so total LED current is 5A for five strips.
LED_IN1
VDD_5V
C24
C25
1uF
0.1uF
GND
LED_IN2
VDD_5V
C26
C27
1uF
0.1uF
GND
LED_IN3
VDD_5V
C28
C29
1uF
0.1uF
GND
LED_IN4
VDD_5V
C30
C31
1uF
0.1uF
GND
LED_IN5
VDD_5V
C32
C33
1uF
0.1uF
GND
VDD_5V
U10
3
VIN
VOUT
1
LED_IN1
LED_EN1
4
EN
GND
2
MIC94082YFT-TR
GND
VDD_5V
U11
3
VIN
VOUT
1
LED_IN2
LED_EN2
4
EN
GND
2
MIC94082YFT-TR
GND
VDD_5V
U13
3
VIN
VOUT
1
LED_IN3
LED_EN3
4
EN
GND
2
MIC94082YFT-TR
GND
VDD_5V
U9
3
VIN
VOUT
1
LED_IN4
LED_EN4
4
EN
GND
2
MIC94082YFT-TR
GND
VDD_5V
U12
3
VIN
VOUT
1
LED_IN5
LED_EN5
4
EN
GND
2
MIC94082YFT-TR
GND
VDD_5V
LED_IN1
LED_IN2
LED_IN3
LED_IN4
LED_IN5
VDD_5V
LED_IN1
LED_IN2
LED_IN3
LED_IN4
LED_IN5
GND
GND
LED Power Switches


USB
USB_DM
USB_DP
MCU_IN
MCU_IN1
MCU_IN2
MCU_IN3
MCU_IN4
LED_EN
LED_EN1
LED_EN2
LED_EN3
LED_EN4
LED_EN5
PWM_SIG
PWM_1
PWM_2
PWM_3
PWM_4
PWM_5
SPI
SPI_MISO
SPI_MOSI
SPI_CS
SPI_CLK
VCC_3V3
C6 0.1uF
C7 0.1uF
C13 0.1uF
C14 0.1uF
C15 1uF
C16 4.7uF
C17 10000pF
GND
RST
SW1
B3U-1000P
C18 0.1uF
U2A
VDD
VDDA
PC13
NRST
BOOT0
PC14-OSC32_IN
PC15-OSC32_OUT
PF0-OSC_IN
PF1-OSC_OUT
VSSA
VSS
X32_IN
X32_OUT
XIN
XOUT
STM32F070CBT6TR
U2B
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
R23 10K PB6
SW3
B3U-1000P
C34 0.1uF
Main Controller
SW2
R4 10K BOOT0
SS-3235D-03-L2
X1
32.768 KHz
C20 10pF
C21 10pF
X2
C22 20pF
C23 20pF
VCC_3V3
GND



Optocoupler Section


PWM_SIG
PWM_1
PWM_2
PWM_3
PWM_4
PWM_5
SPI
SPI_MISO
SPI_MOSI
SPI_CS
SPI_CLK
BUTTONS
BUTTON_1
BUTTON_2
BUTTON_3
BUTTON_4
USB
USB_DM
USB_DP
VBUS_5V
VCC_3V3
R6
10K
R21
36K
Q1
MMBT2222A-7-F
R22
1.5K USB_DP
GND
J8
MH1
MH2
MH3
MH4
MP1
MP2
10118194-0001LF
USB Programming
VDD_5V
C10
0.1uF
C11
1uF
C12
0.1uF
VBAT
BUTTON_2 D5
BUTTON_1
BUTTON_4 D6
BUTTON_3
J3
WJ126V-5.0-6P
Power In
U8
VCCA
VCCB
OE
A1
A2
A3
A4
A5
A6
A7
A8
DIR
B1
B2
B3
B4
B5
B6
B7
B8
R19
R20
SN74LVC8T245PWR
Level translator
J6
Supply Voltage
Date Line {Bit 0}
Date Line {Bit 1}
Date Line {Bit 2}
Card Det/Date line{Bit 3}
Command/Response
Clock
MP4
MP3
MP2
MP1
Supply Voltage GND
TF-002T-15
SD Card Connector
LED_IN1
LED_IN2
LED_IN3
LED_IN4
LED_IN5
#PWM_1
#PWM_2
#PWM_3
#PWM_4
#PWM_5
J1
J2
J4
J5
J7
640456-3
LED Out